

PART A – Probability & Conceptual Foundation (Theory)

Q1. What is Conditional Probability?

Answer:

Conditional probability is the probability of an event happening given that another event has already happened. It's written as $P(A|B)$, meaning the probability of A occurring given that B is true. For example, the probability that a message is spam given that it contains a URL is a conditional probability, it narrows the sample space down to only messages that already satisfy the given condition.

Q2. Explain Bayes' Theorem and its importance in classification problems.

Answer:

Bayes' Theorem describes how to update the probability of an event based on new evidence. It is written as $P(A|B) = [P(B|A) * P(A)] / P(B)$. In classification, this lets us calculate the probability of a class (like Spam) given the observed features of a message, by combining how likely those features are for that class with how common that class is overall. This is the core idea behind probabilistic classifiers like Naive Bayes, since it gives a mathematical way to go from "what features does this message have" to "what class does it most likely belong to."

Q3. What assumptions does the Naive Bayes Classifier make?

Answer:

Naive Bayes assumes that all the input features are conditionally independent of each other given the class label. In other words, it assumes that knowing the value of one feature (like num_urls) tells us nothing extra about another feature (like spam_keyword_score) once we already know the class. This assumption is rarely fully true in real data since features are often related to some degree, which is why it's called "naive", but the classifier still tends to work surprisingly well in practice despite this simplification.

Q4. Explain the working principle of K-Nearest Neighbors (KNN) and Support Vector Machine (SVM).

Answer:

K-Nearest Neighbors classifies a new data point by looking at the 'k' closest points to it in the training data (based on a distance metric like Euclidean distance) and assigning the majority

class among those neighbors. It doesn't really "learn" a model in the traditional sense, it just stores the training data and compares new points against it at prediction time. Support Vector Machine, on the other hand, tries to find the best possible boundary (called a hyperplane) that separates the two classes with the maximum possible margin between them. The points closest to this boundary, called support vectors, are what actually decide where the boundary goes. SVM can also use kernels (like RBF or polynomial) to handle cases where the classes aren't separable by a simple straight line.

Q5. Compare distance-based, probabilistic, and margin-based classifiers.

Answer:

Distance-based classifiers like KNN make predictions purely based on how close a new point is to existing labeled points, with no real assumption about the underlying data distribution. Probabilistic classifiers like Naive Bayes work by estimating the probability of each class given the features and picking the class with the higher probability, which also lets us know how confident the prediction is. Margin-based classifiers like SVM focus on finding the decision boundary that keeps the maximum gap (margin) between the two classes, prioritizing a boundary that generalizes well rather than just fitting the training points closely. In short, KNN cares about proximity, Naive Bayes cares about probability, and SVM cares about the cleanest possible separation between classes.